# COVID-19 Data Analysis

# 📌 Step 1: Title & Introduction

### COVID-19 Data Analysis using Python 📊

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**Dataset Source:** COVID-19 Dataset
https://www.kaggle.com/datasets/meirnizri/covid19-dataset

## 📌 Project Overview

This project analyzes the **COVID-19 dataset** to uncover insights into global trends, case spikes, mortality rates, and vaccination progress.
We perform **data preprocessing, exploratory analysis (EDA), and visualization** to understand key patterns.

# 📌 Step 2: Importing Necessary Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from scipy.stats import zscore
import math
from scipy.stats.mstats import winsorize
```

# 📌 Step 3: Loading & Understanding the Dataset

```
from google.colab import drive
drive.mount('/content/drive')
df = pd.read_csv('/content/drive/My Drive/Data Sets/covid-data.csv')
```

```
Mounted at /content/drive
```

**Note**: Some the output of notebook does not present the complete output, therefore we can increase the limit of columns view and row view by using these commands:

```
pd.set_option('display.max_columns', None) # this is to display all the columns in the dataframe
pd.set_option('display.max_rows', None) # this is to display all the rows in the dataframe

# hide all warnings runtime
import warnings
warnings.filterwarnings('ignore')
```

# Step 4: Data Summary & Initial Cleaning

```
# Display the first few rows
df.head()
```

{"type":"dataframe","variable_name":"df"}

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 302512 entries, 0 to 302511
Data columns (total 67 columns):
 #   Column                                      Non-Null Count
Dtype
---  ------                                      --------------
-----
 0   iso_code                                    302512 non-null
object
 1   continent                                   288160 non-null
object
 2   location                                    302512 non-null
object
 3   date                                        302512 non-null
object
 4   total_cases                                 266771 non-null
float64
 5   new_cases                                   294064 non-null
float64
 6   new_cases_smoothed                          292800 non-null
float64
 7   total_deaths                                246214 non-null
float64
 8   new_deaths                                  294139 non-null
float64
```

```
 9   new_deaths_smoothed                          292909 non-null
float64
 10  total_cases_per_million                      266771 non-null
float64
 11  new_cases_per_million                        294064 non-null
float64
 12  new_cases_smoothed_per_million               292800 non-null
float64
 13  total_deaths_per_million                     246214 non-null
float64
 14  new_deaths_per_million                       294139 non-null
float64
 15  new_deaths_smoothed_per_million              292909 non-null
float64
 16  reproduction_rate                            184817 non-null
float64
 17  icu_patients                                 34764 non-null
float64
 18  icu_patients_per_million                     34764 non-null
float64
 19  hosp_patients                                35138 non-null
float64
 20  hosp_patients_per_million                    35138 non-null
float64
 21  weekly_icu_admissions                        9101 non-null
float64
 22  weekly_icu_admissions_per_million            9101 non-null
float64
 23  weekly_hosp_admissions                       21287 non-null
float64
 24  weekly_hosp_admissions_per_million           21287 non-null
float64
 25  total_tests                                  79387 non-null
float64
 26  new_tests                                    75403 non-null
float64
 27  total_tests_per_thousand                     79387 non-null
float64
 28  new_tests_per_thousand                       75403 non-null
float64
 29  new_tests_smoothed                           103965 non-null
float64
 30  new_tests_smoothed_per_thousand              103965 non-null
float64
 31  positive_rate                                95927 non-null
float64
 32  tests_per_case                               94348 non-null
float64
 33  tests_units                                  106788 non-null
```

```
object
 34  total_vaccinations                          73561 non-null
float64
 35  people_vaccinated                           70411 non-null
float64
 36  people_fully_vaccinated                     68149 non-null
float64
 37  total_boosters                              42324 non-null
float64
 38  new_vaccinations                            60542 non-null
float64
 39  new_vaccinations_smoothed                   163536 non-null
float64
 40  total_vaccinations_per_hundred              73561 non-null
float64
 41  people_vaccinated_per_hundred               70411 non-null
float64
 42  people_fully_vaccinated_per_hundred         68149 non-null
float64
 43  total_boosters_per_hundred                  42324 non-null
float64
 44  new_vaccinations_smoothed_per_million       163536 non-null
float64
 45  new_people_vaccinated_smoothed              163587 non-null
float64
 46  new_people_vaccinated_smoothed_per_hundred  163587 non-null
float64
 47  stringency_index                            193194 non-null
float64
 48  population_density                          256703 non-null
float64
 49  median_age                                  238751 non-null
float64
 50  aged_65_older                               230391 non-null
float64
 51  aged_70_older                               236359 non-null
float64
 52  gdp_per_capita                              233979 non-null
float64
 53  extreme_poverty                             150700 non-null
float64
 54  cardiovasc_death_rate                       234406 non-null
float64
 55  diabetes_prevalence                         246348 non-null
float64
 56  female_smokers                              175815 non-null
float64
 57  male_smokers                                173423 non-null
float64
```

```
 58  handwashing_facilities                         114817 non-null
float64
 59  hospital_beds_per_thousand                     206911 non-null
float64
 60  life_expectancy                                278219 non-null
float64
 61  human_development_index                        227212 non-null
float64
 62  population                                     302512 non-null
float64
 63  excess_mortality_cumulative_absolute           10295 non-null
float64
 64  excess_mortality_cumulative                    10295 non-null
float64
 65  excess_mortality                               10295 non-null
float64
 66  excess_mortality_cumulative_per_million        10295 non-null
float64
dtypes: float64(62), object(5)
memory usage: 154.6+ MB
```

# # Observations

1. There are 302512 rows and 67 columns in the dataset
2. The columns are of different data types
3. The columns in the datasets are:
   - 'iso_code', 'continent', 'location', 'date', 'total_cases', 'new_cases', 'new_cases_smoothed', 'total_deaths', 'new_deaths', 'new_deaths_smoothed', 'total_cases_per_million', 'new_cases_per_million', 'new_cases_smoothed_per_million', 'total_deaths_per_million', 'new_deaths_per_million', 'new_deaths_smoothed_per_million', 'reproduction_rate', 'icu_patients', 'icu_patients_per_million', 'hosp_patients', 'hosp_patients_per_million', 'weekly_icu_admissions', 'weekly_icu_admissions_per_million', 'weekly_hosp_admissions', 'weekly_hosp_admissions_per_million', 'total_tests', 'new_tests', 'total_tests_per_thousand', 'new_tests_per_thousand', 'new_tests_smoothed', 'new_tests_smoothed_per_thousand', 'positive_rate', 'tests_per_case', 'tests_units', 'total_vaccinations', 'people_vaccinated', 'people_fully_vaccinated', 'total_boosters', 'new_vaccinations', 'new_vaccinations_smoothed', 'total_vaccinations_per_hundred', 'people_vaccinated_per_hundred', 'people_fully_vaccinated_per_hundred', 'total_boosters_per_hundred', 'new_vaccinations_smoothed_per_million', 'new_people_vaccinated_smoothed', 'new_people_vaccinated_smoothed_per_hundred', 'stringency_index', 'population_density', 'median_age', 'aged_65_older', 'aged_70_older', 'gdp_per_capita', 'extreme_poverty', 'cardiovasc_death_rate', 'diabetes_prevalence', 'female_smokers', 'male_smokers', 'handwashing_facilities', 'hospital_beds_per_thousand', 'life_expectancy', 'human_development_index', 'population',

'excess_mortality_cumulative_absolute', 'excess_mortality_cumulative', 'excess_mortality', 'excess_mortality_cumulative_per_million'

4. There are some missing values in the dataset which we will read in details and deal later on in the notebook.

---

```
df.sample(50)
```

{"type":"dataframe"}

- `df.sample(50)` gives us whole picture or idea about data.

```
df.columns

Index(['iso_code', 'continent', 'location', 'date', 'total_cases',
'new_cases',
       'new_cases_smoothed', 'total_deaths', 'new_deaths',
       'new_deaths_smoothed', 'total_cases_per_million',
       'new_cases_per_million', 'new_cases_smoothed_per_million',
       'total_deaths_per_million', 'new_deaths_per_million',
       'new_deaths_smoothed_per_million', 'reproduction_rate',
'icu_patients',
       'icu_patients_per_million', 'hosp_patients',
       'hosp_patients_per_million', 'weekly_icu_admissions',
       'weekly_icu_admissions_per_million', 'weekly_hosp_admissions',
       'weekly_hosp_admissions_per_million', 'total_tests',
'new_tests',
       'total_tests_per_thousand', 'new_tests_per_thousand',
       'new_tests_smoothed', 'new_tests_smoothed_per_thousand',
       'positive_rate', 'tests_per_case', 'tests_units',
'total_vaccinations',
       'people_vaccinated', 'people_fully_vaccinated',
'total_boosters',
       'new_vaccinations', 'new_vaccinations_smoothed',
       'total_vaccinations_per_hundred',
'people_vaccinated_per_hundred',
       'people_fully_vaccinated_per_hundred',
'total_boosters_per_hundred',
       'new_vaccinations_smoothed_per_million',
       'new_people_vaccinated_smoothed',
       'new_people_vaccinated_smoothed_per_hundred',
'stringency_index',
       'population_density', 'median_age', 'aged_65_older',
'aged_70_older',
       'gdp_per_capita', 'extreme_poverty', 'cardiovasc_death_rate',
       'diabetes_prevalence', 'female_smokers', 'male_smokers',
       'handwashing_facilities', 'hospital_beds_per_thousand',
       'life_expectancy', 'human_development_index', 'population',
       'excess_mortality_cumulative_absolute',
'excess_mortality_cumulative',
```

```
       'excess_mortality', 'excess_mortality_cumulative_per_million'],
      dtype='object')
```

- let's see the exact column names which can be easily copied later

```
# Descriptive statistics
df.describe(include='all')
```

{"type":"dataframe"}

# 🧹 Step 5: Handling Missing Values

```
# Check for missing values
df.isnull().sum().sort_values(ascending=False) # this will show the
number of null values in each column in descending order
```

```
weekly_icu_admissions                          293411
weekly_icu_admissions_per_million              293411
excess_mortality_cumulative_absolute           292217
excess_mortality_cumulative_per_million        292217
excess_mortality_cumulative                    292217
excess_mortality                               292217
weekly_hosp_admissions_per_million             281225
weekly_hosp_admissions                         281225
icu_patients_per_million                       267748
icu_patients                                   267748
hosp_patients_per_million                      267374
hosp_patients                                  267374
total_boosters                                 260188
total_boosters_per_hundred                     260188
new_vaccinations                               241970
people_fully_vaccinated                        234363
people_fully_vaccinated_per_hundred            234363
people_vaccinated                              232101
people_vaccinated_per_hundred                  232101
total_vaccinations_per_hundred                 228951
total_vaccinations                             228951
new_tests                                      227109
new_tests_per_thousand                         227109
total_tests                                    223125
total_tests_per_thousand                       223125
tests_per_case                                 208164
positive_rate                                  206585
new_tests_smoothed_per_thousand                198547
new_tests_smoothed                             198547
tests_units                                    195724
handwashing_facilities                         187695
extreme_poverty                                151812
```

```
new_vaccinations_smoothed_per_million         138976
new_vaccinations_smoothed                     138976
new_people_vaccinated_smoothed_per_hundred    138925
new_people_vaccinated_smoothed                138925
male_smokers                                  129089
female_smokers                                126697
reproduction_rate                             117695
stringency_index                              109318
hospital_beds_per_thousand                     95601
human_development_index                        75300
aged_65_older                                  72121
gdp_per_capita                                 68533
cardiovasc_death_rate                          68106
aged_70_older                                  66153
median_age                                     63761
total_deaths                                   56298
total_deaths_per_million                       56298
diabetes_prevalence                            56164
population_density                             45809
total_cases_per_million                        35741
total_cases                                    35741
life_expectancy                                24293
continent                                      14352
new_cases_smoothed                              9712
new_cases_smoothed_per_million                  9712
new_deaths_smoothed_per_million                 9603
new_deaths_smoothed                             9603
new_cases                                       8448
new_cases_per_million                           8448
new_deaths                                      8373
new_deaths_per_million                          8373
population                                         0
date                                               0
location                                           0
iso_code                                           0
dtype: int64
```

```python
(df.isnull().sum() / len(df) * 100).sort_values(ascending=False) # this will show the percentage of null values in each column
```

```
weekly_icu_admissions                         96.991524
weekly_icu_admissions_per_million             96.991524
excess_mortality_cumulative_absolute          96.596829
excess_mortality_cumulative_per_million       96.596829
excess_mortality_cumulative                   96.596829
excess_mortality                              96.596829
weekly_hosp_admissions_per_million            92.963254
weekly_hosp_admissions                        92.963254
icu_patients_per_million                      88.508224
icu_patients                                  88.508224
```

| | |
|---|---|
| hosp_patients_per_million | 88.384593 |
| hosp_patients | 88.384593 |
| total_boosters | 86.009150 |
| total_boosters_per_hundred | 86.009150 |
| new_vaccinations | 79.986910 |
| people_fully_vaccinated | 77.472299 |
| people_fully_vaccinated_per_hundred | 77.472299 |
| people_vaccinated | 76.724560 |
| people_vaccinated_per_hundred | 76.724560 |
| total_vaccinations_per_hundred | 75.683279 |
| total_vaccinations | 75.683279 |
| new_tests | 75.074377 |
| new_tests_per_thousand | 75.074377 |
| total_tests | 73.757405 |
| total_tests_per_thousand | 73.757405 |
| tests_per_case | 68.811816 |
| positive_rate | 68.289853 |
| new_tests_smoothed_per_thousand | 65.632768 |
| new_tests_smoothed | 65.632768 |
| tests_units | 64.699582 |
| handwashing_facilities | 62.045473 |
| extreme_poverty | 50.183794 |
| new_vaccinations_smoothed_per_million | 45.940657 |
| new_vaccinations_smoothed | 45.940657 |
| new_people_vaccinated_smoothed_per_hundred | 45.923798 |
| new_people_vaccinated_smoothed | 45.923798 |
| male_smokers | 42.672357 |
| female_smokers | 41.881644 |
| reproduction_rate | 38.905895 |
| stringency_index | 36.136748 |
| hospital_beds_per_thousand | 31.602383 |
| human_development_index | 24.891575 |
| aged_65_older | 23.840707 |
| gdp_per_capita | 22.654638 |
| cardiovasc_death_rate | 22.513487 |
| aged_70_older | 21.867893 |
| median_age | 21.077180 |
| total_deaths | 18.610171 |
| total_deaths_per_million | 18.610171 |
| diabetes_prevalence | 18.565875 |
| population_density | 15.142870 |
| total_cases_per_million | 11.814738 |
| total_cases | 11.814738 |
| life_expectancy | 8.030425 |
| continent | 4.744275 |
| new_cases_smoothed | 3.210451 |
| new_cases_smoothed_per_million | 3.210451 |
| new_deaths_smoothed_per_million | 3.174420 |
| new_deaths_smoothed | 3.174420 |

```
new_cases                                    2.792616
new_cases_per_million                        2.792616
new_deaths                                   2.767824
new_deaths_per_million                       2.767824
population                                   0.000000
date                                         0.000000
location                                     0.000000
iso_code                                     0.000000
dtype: float64
```

```python
# Forward fill and Backward fill for time-series or categorical data
df.fillna(method='ffill', inplace=True)
df.fillna(method='bfill', inplace=True)

# Separate numerical and categorical columns
num_cols = df.select_dtypes(include=['number']).columns
cat_cols = df.select_dtypes(include=['object']).columns

# Fill numerical columns with median (in case ffill and bfill didn't work)
df[num_cols] = df[num_cols].apply(lambda x: x.fillna(x.median()))

# Fill categorical columns with mode
df[cat_cols] = df[cat_cols].apply(lambda x: x.fillna(x.mode()[0]))

# Check if all missing values are handled
print("Remaining missing values per column:")
print(df.isnull().sum().sum())  # Should be 0
```

```
Remaining missing values per column:
0
```

# Step 6: Identifying & Removing Duplicates

```python
# Check for duplicate rows
duplicates = df.duplicated().sum()
print(f"Number of duplicate rows: {duplicates}")
```

```
Number of duplicate rows: 0
```

```python
# Look for any columns with invalid data types or strange values
print(df.dtypes)
```

```
iso_code                                     object
continent                                    object
location                                     object
date                                         object
total_cases                                 float64
```

```
new_cases                                  float64
new_cases_smoothed                         float64
total_deaths                               float64
new_deaths                                 float64
new_deaths_smoothed                        float64
total_cases_per_million                    float64
new_cases_per_million                      float64
new_cases_smoothed_per_million             float64
total_deaths_per_million                   float64
new_deaths_per_million                     float64
new_deaths_smoothed_per_million            float64
reproduction_rate                          float64
icu_patients                               float64
icu_patients_per_million                   float64
hosp_patients                              float64
hosp_patients_per_million                  float64
weekly_icu_admissions                      float64
weekly_icu_admissions_per_million          float64
weekly_hosp_admissions                     float64
weekly_hosp_admissions_per_million         float64
total_tests                                float64
new_tests                                  float64
total_tests_per_thousand                   float64
new_tests_per_thousand                     float64
new_tests_smoothed                         float64
new_tests_smoothed_per_thousand            float64
positive_rate                              float64
tests_per_case                             float64
tests_units                                 object
total_vaccinations                         float64
people_vaccinated                          float64
people_fully_vaccinated                    float64
total_boosters                             float64
new_vaccinations                           float64
new_vaccinations_smoothed                  float64
total_vaccinations_per_hundred             float64
people_vaccinated_per_hundred              float64
people_fully_vaccinated_per_hundred        float64
total_boosters_per_hundred                 float64
new_vaccinations_smoothed_per_million      float64
new_people_vaccinated_smoothed             float64
new_people_vaccinated_smoothed_per_hundred float64
stringency_index                           float64
population_density                         float64
median_age                                 float64
aged_65_older                              float64
aged_70_older                              float64
gdp_per_capita                             float64
extreme_poverty                            float64
```

```
cardiovasc_death_rate                                   float64
diabetes_prevalence                                     float64
female_smokers                                          float64
male_smokers                                            float64
handwashing_facilities                                  float64
hospital_beds_per_thousand                              float64
life_expectancy                                         float64
human_development_index                                 float64
population                                              float64
excess_mortality_cumulative_absolute                    float64
excess_mortality_cumulative                             float64
excess_mortality                                        float64
excess_mortality_cumulative_per_million                 float64
dtype: object
```

## 🧩 Step 7: Checking for Inconsistent Categorical Values

```
# Identify categorical columns
cat_cols = df.select_dtypes(include=['object']).columns

# Check for inconsistent categorical values
for col in cat_cols:
    print(f"Unique values in '{col}':")
    print(df[col].value_counts(dropna=False))  # Includes NaN counts
    print("-" * 50)  # Separator for readability

Unique values in 'iso_code':
iso_code
ARG            1198
MEX            1198
AFG            1196
PLW            1196
NIC            1196
NER            1196
NGA            1196
NIU            1196
OWID_NAM       1196
PRK            1196
MKD            1196
MNP            1196
NOR            1196
OWID_OCE       1196
OMN            1196
PAK            1196
PSE            1196
NCL            1196
```

| | |
|---|---|
| PAN | 1196 |
| PNG | 1196 |
| PRY | 1196 |
| PER | 1196 |
| PHL | 1196 |
| PCN | 1196 |
| POL | 1196 |
| PRT | 1196 |
| PRI | 1196 |
| QAT | 1196 |
| REU | 1196 |
| ROU | 1196 |
| RUS | 1196 |
| NZL | 1196 |
| NLD | 1196 |
| BLM | 1196 |
| MRT | 1196 |
| LTU | 1196 |
| OWID_AFR | 1196 |
| OWID_LMC | 1196 |
| LUX | 1196 |
| MDG | 1196 |
| MWI | 1196 |
| MYS | 1196 |
| MDV | 1196 |
| MLI | 1196 |
| MLT | 1196 |
| MHL | 1196 |
| MTQ | 1196 |
| MUS | 1196 |
| NPL | 1196 |
| MYT | 1196 |
| FSM | 1196 |
| MDA | 1196 |
| MCO | 1196 |
| MNG | 1196 |
| MNE | 1196 |
| MSR | 1196 |
| MAR | 1196 |
| MOZ | 1196 |
| MMR | 1196 |
| NAM | 1196 |
| NRU | 1196 |
| RWA | 1196 |
| SHN | 1196 |
| LBY | 1196 |
| TZA | 1196 |
| TLS | 1196 |
| TGO | 1196 |

```
TKL         1196
TON         1196
TTO         1196
TUN         1196
TUR         1196
TKM         1196
TCA         1196
TUV         1196
UGA         1196
UKR         1196
ARE         1196
GBR         1196
USA         1196
VIR         1196
OWID_UMC    1196
URY         1196
UZB         1196
VUT         1196
VAT         1196
VEN         1196
VNM         1196
WLF         1196
OWID_WRL    1196
YEM         1196
ZMB         1196
THA         1196
TJK         1196
KNA         1196
SYR         1196
LCA         1196
MAF         1196
SPM         1196
VCT         1196
WSM         1196
SMR         1196
STP         1196
SAU         1196
SEN         1196
SRB         1196
SYC         1196
SLE         1196
SGP         1196
SXM         1196
SVK         1196
SVN         1196
SLB         1196
SOM         1196
ZAF         1196
OWID_SAM    1196
```

```
KOR         1196
SSD         1196
ESP         1196
LKA         1196
SDN         1196
SWE         1196
CHE         1196
LIE         1196
OWID_LIC    1196
LBR         1196
COG         1196
BFA         1196
BDI         1196
KHM         1196
CMR         1196
CAN         1196
CPV         1196
CYM         1196
CAF         1196
TCD         1196
CHL         1196
CHN         1196
COL         1196
COM         1196
COK         1196
SLV         1196
CRI         1196
CIV         1196
HRV         1196
CUB         1196
CUW         1196
CYP         1196
CZE         1196
COD         1196
DNK         1196
DJI         1196
DMA         1196
DOM         1196
LSO         1196
BGR         1196
BRN         1196
VGB         1196
BRA         1196
ALB         1196
DZA         1196
ASM         1196
AND         1196
AGO         1196
AIA         1196
```

| | |
|---|---|
| ATG | 1196 |
| ARM | 1196 |
| ABW | 1196 |
| OWID_ASI | 1196 |
| AUS | 1196 |
| AUT | 1196 |
| AZE | 1196 |
| BHS | 1196 |
| BHR | 1196 |
| BGD | 1196 |
| BRB | 1196 |
| BLR | 1196 |
| BEL | 1196 |
| BLZ | 1196 |
| BEN | 1196 |
| BMU | 1196 |
| BTN | 1196 |
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North America    52626
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Niue                                        1196
North America                               1196
North Korea                                 1196
North Macedonia                             1196
Northern Mariana Islands                    1196
Norway                                      1196
Oceania                                     1196
Oman                                        1196
Pakistan                                    1196
Palestine                                   1196
New Caledonia                               1196
Panama                                      1196
Papua New Guinea                            1196
Paraguay                                    1196
Peru                                        1196
Philippines                                 1196
Pitcairn                                    1196
Poland                                      1196
Portugal                                    1196
Puerto Rico                                 1196
Qatar                                       1196
Reunion                                     1196
Romania                                     1196
Russia                                      1196
New Zealand                                 1196
Netherlands                                 1196
Saint Barthelemy                            1196
Mauritania                                  1196
Lithuania                                   1196
Africa                                      1196
Lower middle income                         1196
Luxembourg                                  1196
Madagascar                                  1196
Malawi                                      1196
Malaysia                                    1196
Maldives                                    1196
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| Marshall Islands | 1196 |
| Martinique | 1196 |
| Mauritius | 1196 |
| Nepal | 1196 |
| Mayotte | 1196 |
| Micronesia (country) | 1196 |
| Moldova | 1196 |
| Monaco | 1196 |
| Mongolia | 1196 |
| Montenegro | 1196 |
| Montserrat | 1196 |
| Morocco | 1196 |
| Mozambique | 1196 |
| Myanmar | 1196 |
| Namibia | 1196 |
| Nauru | 1196 |
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| Saint Helena | 1196 |
| Libya | 1196 |
| Tanzania | 1196 |
| Timor | 1196 |
| Togo | 1196 |
| Tokelau | 1196 |
| Tonga | 1196 |
| Trinidad and Tobago | 1196 |
| Tunisia | 1196 |
| Turkey | 1196 |
| Turkmenistan | 1196 |
| Turks and Caicos Islands | 1196 |
| Tuvalu | 1196 |
| Uganda | 1196 |
| Ukraine | 1196 |
| United Arab Emirates | 1196 |
| United Kingdom | 1196 |
| United States | 1196 |
| United States Virgin Islands | 1196 |
| Upper middle income | 1196 |
| Uruguay | 1196 |
| Uzbekistan | 1196 |
| Vanuatu | 1196 |
| Vatican | 1196 |
| Venezuela | 1196 |
| Vietnam | 1196 |
| Wallis and Futuna | 1196 |
| World | 1196 |
| Yemen | 1196 |
| Zambia | 1196 |

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| Tajikistan | 1196 |
| Saint Kitts and Nevis | 1196 |
| Syria | 1196 |
| Saint Lucia | 1196 |
| Saint Martin (French part) | 1196 |
| Saint Pierre and Miquelon | 1196 |
| Saint Vincent and the Grenadines | 1196 |
| Samoa | 1196 |
| San Marino | 1196 |
| Sao Tome and Principe | 1196 |
| Saudi Arabia | 1196 |
| Senegal | 1196 |
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| Seychelles | 1196 |
| Sierra Leone | 1196 |
| Singapore | 1196 |
| Sint Maarten (Dutch part) | 1196 |
| Slovakia | 1196 |
| Slovenia | 1196 |
| Solomon Islands | 1196 |
| Somalia | 1196 |
| South Africa | 1196 |
| South America | 1196 |
| South Korea | 1196 |
| South Sudan | 1196 |
| Spain | 1196 |
| Sri Lanka | 1196 |
| Sudan | 1196 |
| Sweden | 1196 |
| Switzerland | 1196 |
| Liechtenstein | 1196 |
| Low income | 1196 |
| Liberia | 1196 |
| Congo | 1196 |
| Burkina Faso | 1196 |
| Burundi | 1196 |
| Cambodia | 1196 |
| Cameroon | 1196 |
| Canada | 1196 |
| Cape Verde | 1196 |
| Cayman Islands | 1196 |
| Central African Republic | 1196 |
| Chad | 1196 |
| Chile | 1196 |
| China | 1196 |
| Colombia | 1196 |
| Comoros | 1196 |
| Cook Islands | 1196 |

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| Cote d'Ivoire | 1196 |
| Croatia | 1196 |
| Cuba | 1196 |
| Curacao | 1196 |
| Cyprus | 1196 |
| Czechia | 1196 |
| Democratic Republic of Congo | 1196 |
| Denmark | 1196 |
| Djibouti | 1196 |
| Dominica | 1196 |
| Dominican Republic | 1196 |
| Lesotho | 1196 |
| Bulgaria | 1196 |
| Brunei | 1196 |
| British Virgin Islands | 1196 |
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| Albania | 1196 |
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| American Samoa | 1196 |
| Andorra | 1196 |
| Angola | 1196 |
| Anguilla | 1196 |
| Antigua and Barbuda | 1196 |
| Armenia | 1196 |
| Aruba | 1196 |
| Asia | 1196 |
| Australia | 1196 |
| Austria | 1196 |
| Azerbaijan | 1196 |
| Bahamas | 1196 |
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| Bermuda | 1196 |
| Bhutan | 1196 |
| Bolivia | 1196 |
| Bonaire Sint Eustatius and Saba | 1196 |
| Bosnia and Herzegovina | 1196 |
| Botswana | 1196 |
| Egypt | 1196 |
| Ecuador | 1196 |
| Zimbabwe | 1196 |
| Indonesia | 1196 |

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| Guinea | 1196 |
| Equatorial Guinea | 1196 |
| Guinea-Bissau | 1196 |
| Guyana | 1196 |
| Haiti | 1196 |
| High income | 1196 |
| Honduras | 1196 |
| Kyrgyzstan | 1196 |
| Hungary | 1196 |
| Iceland | 1196 |
| India | 1196 |
| Iran | 1196 |
| Kuwait | 1196 |
| Iraq | 1196 |
| Ireland | 1196 |
| Isle of Man | 1196 |
| Israel | 1196 |
| Italy | 1196 |
| Jamaica | 1196 |
| Japan | 1196 |
| Jersey | 1196 |
| Jordan | 1196 |
| Kazakhstan | 1196 |
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| Guam | 1196 |
| Guadeloupe | 1196 |
| Eritrea | 1196 |
| Estonia | 1196 |
| Eswatini | 1196 |
| Ethiopia | 1196 |
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| European Union | 1196 |
| Faeroe Islands | 1196 |
| Falkland Islands | 1196 |
| Fiji | 1196 |
| Lebanon | 1196 |
| Finland | 1196 |
| France | 1196 |
| French Guiana | 1196 |
| French Polynesia | 1196 |
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| Ghana | 1196 |
| Gibraltar | 1196 |

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Laos                                      1196
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Greenland                                 1196
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Suriname                                  1195
Taiwan                                    1183
Hong Kong                                 1165
Northern Ireland                          1131
Scotland                                  1123
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Macao                                      787
Northern Cyprus                            691
Western Sahara                               1
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2020-01-06    246
2020-01-05    246
2020-01-01      2
2020-01-02      2
Name: count, dtype: int64
--------------------------------------------------
Unique values in 'tests_units':
tests_units
tests performed    223167
people tested       52600
samples tested      25520
units unclear        1225
Name: count, dtype: int64
--------------------------------------------------
```

## Explanation:

- **Identifies categorical columns**
- **Displays all unique values along with their counts**
- **Includes NaN values for completeness**

```
# Identify unique values in 'iso_code'
invalid_iso_codes = ['OWID_CYN', 'ESH']  # These are outliers

# Filter out invalid iso_codes
df = df[~df['iso_code'].isin(invalid_iso_codes)]

# Identify inconsistent location values (e.g., "Lower middle income",
"Africa")
valid_locations = df['location'].value_counts().index.tolist()

# Define a function to check if a location is valid
def is_valid_location(loc):
    invalid_terms = ["income", "World", "continent", "region"]  #
Broad terms that indicate issues
    return not any(term.lower() in loc.lower() for term in
invalid_terms)

# Apply filtering
df = df[df['location'].apply(is_valid_location)]

# Print cleaned categorical data summary
for col in cat_cols:
    print(f"Column: {col}")
    print(df[col].value_counts(dropna=False))
    print("-" * 50)
```

```
Column: iso_code
iso_code
ARG         1198
MEX         1198
AFG         1196
PSE         1196
NER         1196
NGA         1196
NIU         1196
OWID_NAM    1196
PRK         1196
MKD         1196
MNP         1196
NOR         1196
OWID_OCE    1196
OMN         1196
PAK         1196
PLW         1196
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| PAN | 1196 |
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| PNG | 1196 |
| PRY | 1196 |
| PER | 1196 |
| PHL | 1196 |
| PCN | 1196 |
| POL | 1196 |
| PRT | 1196 |
| PRI | 1196 |
| QAT | 1196 |
| REU | 1196 |
| ROU | 1196 |
| RUS | 1196 |
| NIC | 1196 |
| NLD | 1196 |
| NCL | 1196 |
| MUS | 1196 |
| OWID_AFR | 1196 |
| LTU | 1196 |
| LUX | 1196 |
| MDG | 1196 |
| MWI | 1196 |
| MYS | 1196 |
| MDV | 1196 |
| MLI | 1196 |
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| Saint Martin (French part) | 1196 |
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| 2020-03-18 | 245 |
| 2020-03-19 | 245 |
| 2020-03-12 | 245 |
| 2020-03-04 | 244 |
| 2020-03-01 | 244 |
| 2020-03-06 | 244 |
| 2020-03-05 | 244 |
| 2020-03-03 | 244 |
| 2020-03-02 | 244 |
| 2020-02-29 | 243 |
| 2020-02-02 | 243 |
| 2020-02-07 | 243 |
| 2020-02-06 | 243 |
| 2020-02-05 | 243 |
| 2020-02-04 | 243 |
| 2020-02-03 | 243 |
| 2020-01-31 | 243 |
| 2020-02-28 | 243 |
| 2020-02-09 | 243 |
| 2023-04-06 | 243 |
| 2023-04-07 | 243 |
| 2023-04-08 | 243 |
| 2023-04-09 | 243 |
| 2020-02-08 | 243 |
| 2020-02-01 | 243 |
| 2020-02-10 | 243 |
| 2020-02-11 | 243 |
| 2020-02-27 | 243 |
| 2020-02-26 | 243 |
| 2020-02-25 | 243 |
| 2020-02-24 | 243 |
| 2020-02-23 | 243 |
| 2020-02-22 | 243 |
| 2020-02-21 | 243 |
| 2020-02-20 | 243 |
| 2020-02-12 | 243 |
| 2020-02-13 | 243 |
| 2020-02-14 | 243 |
| 2020-02-15 | 243 |
| 2020-02-16 | 243 |
| 2020-02-17 | 243 |
| 2020-02-18 | 243 |
| 2020-02-19 | 243 |
| 2020-01-28 | 242 |
| 2020-01-30 | 242 |
| 2023-04-12 | 242 |
| 2023-04-11 | 242 |
| 2023-04-10 | 242 |
| 2020-01-16 | 242 |

```
2020-01-27    242
2020-01-17    242
2020-01-18    242
2020-01-19    242
2020-01-20    242
2020-01-29    242
2020-01-22    242
2020-01-23    242
2020-01-24    242
2020-01-25    242
2020-01-26    242
2020-01-21    242
2020-01-03    241
2020-01-04    241
2020-01-15    241
2020-01-14    241
2020-01-13    241
2020-01-12    241
2020-01-11    241
2020-01-10    241
2020-01-09    241
2020-01-08    241
2020-01-07    241
2020-01-06    241
2020-01-05    241
2020-01-01      2
2020-01-02      2
Name: count, dtype: int64
--------------------------------------------------
Column: tests_units
tests_units
tests performed    216495
people tested       52600
samples tested      25520
units unclear        1225
Name: count, dtype: int64
--------------------------------------------------
```

To solve inconsistencies and outliers in the categorical data, we applyed the following steps:

## Steps to Clean the Data:

1. **Remove Outliers**:
   - If `OWID_CYN` and `ESH` are not valid `iso_code` values, we can drop them.

2. **Fix Inconsistencies in `location`**:
   - `location` should contain only country/region names, but values like `"Lower middle income"` and `"Africa"` indicate misclassified data.

- We can filter out non-country values.

3. **Handle Missing or Incorrect Values**:
   - If some categories are missing or have typos, we can correct or remove them.

### What This Code Does:

□ **Removes outlier `iso_code` values** (`OWID_CYN` and `ESH`).
□ **Filters out incorrect `location` values** (like `"Lower middle income"`, `"Africa"`, etc.).
□ **Ensures only valid country/region names remain** in `location`.

# □ Step 8: Data Type Conversion & Column Renaming

```
df['date'] = pd.to_datetime(df['date'], errors='coerce')
```

### Explanation:

□ `pd.to_datetime(df['date'])` converts the column to a proper datetime format.
□ `errors='coerce'` ensures that any invalid values (e.g., wrong formats) become `NaT` (Not a Time), preventing errors.

```
# Rename columns
df = df.rename(columns={
    'iso_code': 'country_code',
    'location': 'country',
    'total_cases': 'total_confirmed_cases',
    'new_cases': 'new_confirmed_cases',
    'total_deaths': 'total_deaths_reported',
    'new_deaths': 'new_deaths_reported'
})
```

Rename columns for better readability and consistency.

```
df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 295840 entries, 0 to 302511
Data columns (total 67 columns):
 #   Column                                      Non-Null Count
Dtype
---  ------                                      --------------
-----
 0   country_code                                295840 non-null
object
 1   continent                                   295840 non-null
```

```
object
 2   country                                            295840 non-null
object
 3   date                                               295840 non-null
datetime64[ns]
 4   total_confirmed_cases                              295840 non-null
float64
 5   new_confirmed_cases                                295840 non-null
float64
 6   new_cases_smoothed                                 295840 non-null
float64
 7   total_deaths_reported                              295840 non-null
float64
 8   new_deaths_reported                                295840 non-null
float64
 9   new_deaths_smoothed                                295840 non-null
float64
 10  total_cases_per_million                            295840 non-null
float64
 11  new_cases_per_million                              295840 non-null
float64
 12  new_cases_smoothed_per_million                     295840 non-null
float64
 13  total_deaths_per_million                           295840 non-null
float64
 14  new_deaths_per_million                             295840 non-null
float64
 15  new_deaths_smoothed_per_million                    295840 non-null
float64
 16  reproduction_rate                                  295840 non-null
float64
 17  icu_patients                                       295840 non-null
float64
 18  icu_patients_per_million                           295840 non-null
float64
 19  hosp_patients                                      295840 non-null
float64
 20  hosp_patients_per_million                          295840 non-null
float64
 21  weekly_icu_admissions                              295840 non-null
float64
 22  weekly_icu_admissions_per_million                  295840 non-null
float64
 23  weekly_hosp_admissions                             295840 non-null
float64
 24  weekly_hosp_admissions_per_million                 295840 non-null
float64
 25  total_tests                                        295840 non-null
float64
```

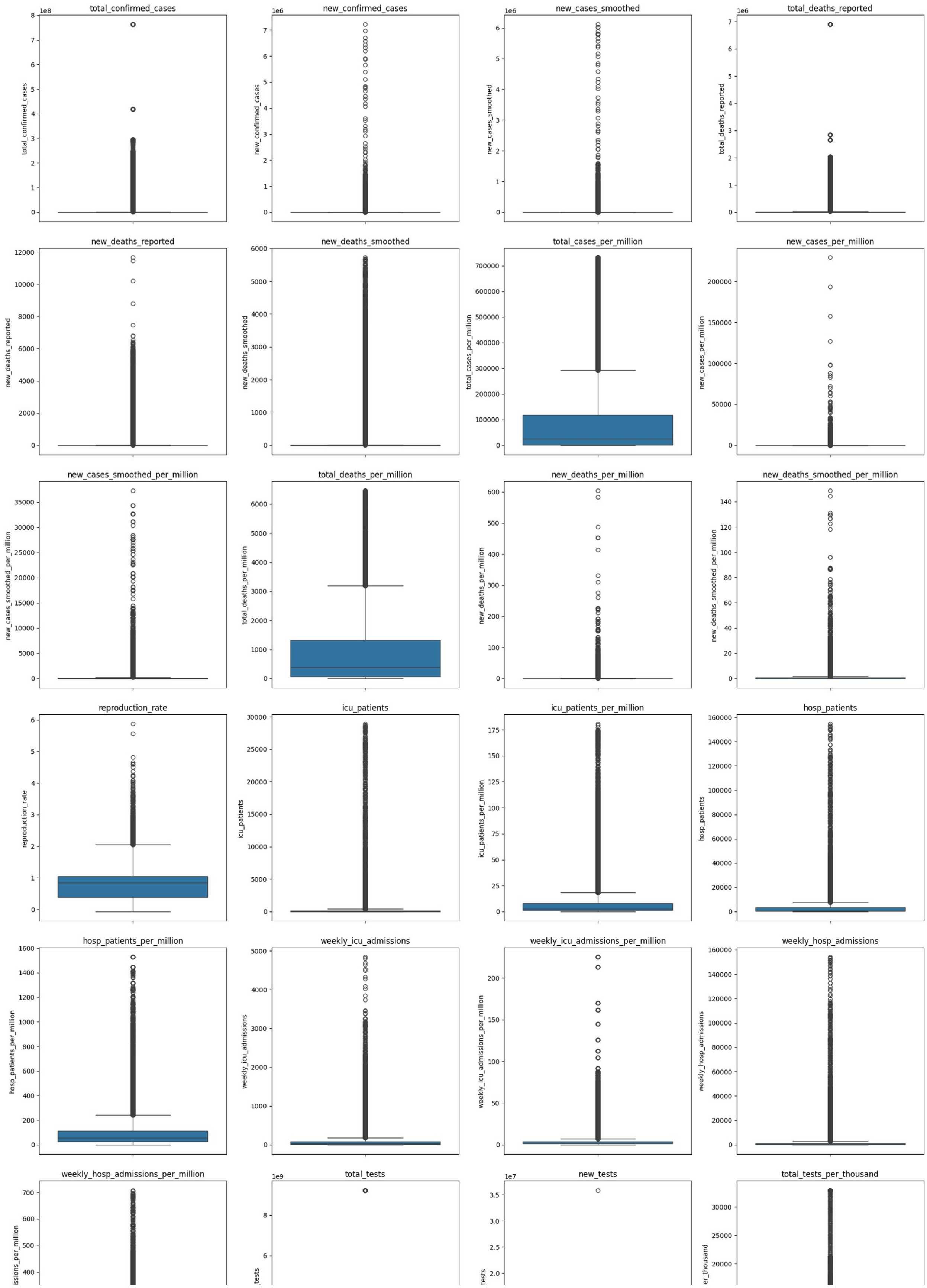
```
 26  new_tests                                  295840 non-null
float64
 27  total_tests_per_thousand                   295840 non-null
float64
 28  new_tests_per_thousand                     295840 non-null
float64
 29  new_tests_smoothed                         295840 non-null
float64
 30  new_tests_smoothed_per_thousand            295840 non-null
float64
 31  positive_rate                              295840 non-null
float64
 32  tests_per_case                             295840 non-null
float64
 33  tests_units                                295840 non-null
object
 34  total_vaccinations                         295840 non-null
float64
 35  people_vaccinated                          295840 non-null
float64
 36  people_fully_vaccinated                    295840 non-null
float64
 37  total_boosters                             295840 non-null
float64
 38  new_vaccinations                           295840 non-null
float64
 39  new_vaccinations_smoothed                  295840 non-null
float64
 40  total_vaccinations_per_hundred             295840 non-null
float64
 41  people_vaccinated_per_hundred              295840 non-null
float64
 42  people_fully_vaccinated_per_hundred        295840 non-null
float64
 43  total_boosters_per_hundred                 295840 non-null
float64
 44  new_vaccinations_smoothed_per_million      295840 non-null
float64
 45  new_people_vaccinated_smoothed             295840 non-null
float64
 46  new_people_vaccinated_smoothed_per_hundred  295840 non-null
float64
 47  stringency_index                           295840 non-null
float64
 48  population_density                         295840 non-null
float64
 49  median_age                                 295840 non-null
float64
 50  aged_65_older                              295840 non-null
```

```
float64
 51  aged_70_older                              295840 non-null
float64
 52  gdp_per_capita                             295840 non-null
float64
 53  extreme_poverty                            295840 non-null
float64
 54  cardiovasc_death_rate                      295840 non-null
float64
 55  diabetes_prevalence                        295840 non-null
float64
 56  female_smokers                             295840 non-null
float64
 57  male_smokers                               295840 non-null
float64
 58  handwashing_facilities                     295840 non-null
float64
 59  hospital_beds_per_thousand                 295840 non-null
float64
 60  life_expectancy                            295840 non-null
float64
 61  human_development_index                    295840 non-null
float64
 62  population                                 295840 non-null
float64
 63  excess_mortality_cumulative_absolute       295840 non-null
float64
 64  excess_mortality_cumulative                295840 non-null
float64
 65  excess_mortality                           295840 non-null
float64
 66  excess_mortality_cumulative_per_million    295840 non-null
float64
dtypes: datetime64[ns](1), float64(62), object(4)
memory usage: 153.5+ MB
```

# 🔍 Step 9: Detecting & Handling Outliers

```
# Select numerical columns
numerical_cols = df.select_dtypes(include=[np.number]).columns
n_cols = len(numerical_cols)
n_rows = math.ceil(n_cols / 4)  # Adjust the number of rows based on
the number of columns

plt.figure(figsize=(20, 5 * n_rows))  # Adjust figure size based on
the number of rows
for i, col in enumerate(numerical_cols, 1):
```

```
    plt.subplot(n_rows, 4, i)
    sns.boxplot(y=df[col])
    plt.title(col)
plt.tight_layout()
plt.show()
```



total_confirmed_cases
1e8
total_confirmed_cases
new_confirmed_cases
1e6
new_confirmed_cases
new_cases_smoothed
1e6
new_cases_smoothed
total_deaths_reported
1e6
total_deaths_reported
new_deaths_reported
new_deaths_reported
new_deaths_smoothed
new_deaths_smoothed
total_cases_per_million
total_cases_per_million
new_cases_per_million
new_cases_per_million
new_cases_smoothed_per_million
new_cases_smoothed_per_million
total_deaths_per_million
total_deaths_per_million
new_deaths_per_million
new_deaths_per_million
new_deaths_smoothed_per_million
new_deaths_smoothed_per_million
reproduction_rate
reproduction_rate
icu_patients
icu_patients
icu_patients_per_million
icu_patients_per_million
hosp_patients
hosp_patients
hosp_patients_per_million
hosp_patients_per_million
weekly_icu_admissions
weekly_icu_admissions
weekly_icu_admissions_per_million
weekly_icu_admissions_per_million
weekly_hosp_admissions
weekly_hosp_admissions
weekly_hosp_admissions_per_million
total_tests
1e9
new_tests
1e7
total_tests_per_thousand

```python
from scipy.stats import zscore

# Calculate Z-scores for numerical columns
z_scores = df[numerical_cols].apply(zscore)

# Define a threshold (e.g., 3 or -3)
threshold = 3
outliers_z = (z_scores > threshold) | (z_scores < -threshold)

# Count outliers in each column
print("Outliers detected using Z-Score:")
print(outliers_z.sum())
```

```
Outliers detected using Z-Score:
total_confirmed_cases                        4269
new_confirmed_cases                          1569
new_cases_smoothed                           1564
total_deaths_reported                        6123
new_deaths_reported                          4001
new_deaths_smoothed                          4106
total_cases_per_million                      6690
new_cases_per_million                        1866
new_cases_smoothed_per_million               3621
total_deaths_per_million                     6286
new_deaths_per_million                       3053
new_deaths_smoothed_per_million              5349
reproduction_rate                            1113
icu_patients                                 2119
icu_patients_per_million                     6180
hosp_patients                                2046
hosp_patients_per_million                    4079
weekly_icu_admissions                        2401
weekly_icu_admissions_per_million            2963
weekly_hosp_admissions                       9758
weekly_hosp_admissions_per_million           3832
total_tests                                   518
new_tests                                    2759
total_tests_per_thousand                     5848
new_tests_per_thousand                       5898
new_tests_smoothed                           1149
new_tests_smoothed_per_thousand              3184
positive_rate                                7901
tests_per_case                                589
total_vaccinations                           3632
people_vaccinated                            3362
people_fully_vaccinated                      3411
total_boosters                               4640
new_vaccinations                             1790
new_vaccinations_smoothed                    1756
total_vaccinations_per_hundred               1058
```

```
people_vaccinated_per_hundred                    0
people_fully_vaccinated_per_hundred              0
total_boosters_per_hundred                    3405
new_vaccinations_smoothed_per_million         6337
new_people_vaccinated_smoothed                1640
new_people_vaccinated_smoothed_per_hundred    5740
stringency_index                                 0
population_density                            4344
median_age                                       0
aged_65_older                                 2392
aged_70_older                                 2392
gdp_per_capita                                5571
extreme_poverty                               2392
cardiovasc_death_rate                         1196
diabetes_prevalence                           4784
female_smokers                                   0
male_smokers                                  2392
handwashing_facilities                           0
hospital_beds_per_thousand                    8372
life_expectancy                                  0
human_development_index                          0
population                                    4784
excess_mortality_cumulative_absolute          8036
excess_mortality_cumulative                   3854
excess_mortality                              5019
excess_mortality_cumulative_per_million       5224
dtype: int64

# Apply winsorization to cap extreme values
df[numerical_cols] = df[numerical_cols].apply(lambda x: winsorize(x,
limits=[0.01, 0.01]))  # Caps top & bottom 1%
```

## Should we Remove Outliers?

**Keep Outliers If:**

- They represent real-world extreme events (e.g., COVID-19 peaks, sudden lockdowns).
- We want to study **rare but important trends** (e.g., record-high death rates).
- Removing them might **hide critical insights** (e.g., the impact of a wave).

**Remove Outliers If:**

- They result from **data entry errors** or anomalies (e.g., negative deaths, impossibly high values).
- They distort **statistical analysis** (e.g., mean, standard deviation).
- We want to **smooth the trend** for forecasting models.

---

## How to Handle Outliers?

If We decide to deal with outliers, here are the best approaches for our dataset:

**1. Winsorization (Capping Outliers)**

- Instead of removing, **cap extreme values** at a percentile threshold (e.g., 99th percentile).
- Best for **keeping trends realistic** while reducing the effect of extreme values.

**2. Transforming Data (Log or Square Root Transform)**

- If data has a **skewed distribution**, applying a **log transform** can reduce the impact of extreme values.

**3. Removing Outliers Based on Z-Score or IQR**

- If extreme values **do not represent real trends**, remove them using **Z-score** or **Interquartile Range (IQR)**.

---

## Best Approach for our Data

Since our dataset includes **real-world COVID-19 data**, the best approach is to **use Winsorization or log transformation rather than direct removal** because:
**1COVID-19 cases & deaths have natural extreme spikes.**
**2 Removing outliers may hide crucial trends (waves, lockdown effects).**
**3 Smoothing extreme values is better than removing them entirely.**

---

```
# Calculate Z-scores for numerical columns
z_scores = df[numerical_cols].apply(zscore)

# Define a threshold (e.g., 3 or -3)
threshold = 3
outliers_z = (z_scores > threshold) | (z_scores < -threshold)

# Count outliers in each column
print("Outliers detected using Z-Score:")
print(outliers_z.sum())
```

```
Outliers detected using Z-Score:
total_confirmed_cases                                5337
new_confirmed_cases                                  6543
new_cases_smoothed                                   6702
total_deaths_reported                                7215
```

| | |
|---|---|
| new_deaths_reported | 6627 |
| new_deaths_smoothed | 6567 |
| total_cases_per_million | 6869 |
| new_cases_per_million | 7602 |
| new_cases_smoothed_per_million | 7674 |
| total_deaths_per_million | 6592 |
| new_deaths_per_million | 8613 |
| new_deaths_smoothed_per_million | 9056 |
| reproduction_rate | 0 |
| icu_patients | 12520 |
| icu_patients_per_million | 7859 |
| hosp_patients | 0 |
| hosp_patients_per_million | 5069 |
| weekly_icu_admissions | 5435 |
| weekly_icu_admissions_per_million | 4564 |
| weekly_hosp_admissions | 10595 |
| weekly_hosp_admissions_per_million | 4741 |
| total_tests | 7934 |
| new_tests | 8230 |
| total_tests_per_thousand | 6546 |
| new_tests_per_thousand | 6563 |
| new_tests_smoothed | 6676 |
| new_tests_smoothed_per_thousand | 7098 |
| positive_rate | 7901 |
| tests_per_case | 4863 |
| total_vaccinations | 5208 |
| people_vaccinated | 4846 |
| people_fully_vaccinated | 4798 |
| total_boosters | 5211 |
| new_vaccinations | 6333 |
| new_vaccinations_smoothed | 6228 |
| total_vaccinations_per_hundred | 0 |
| people_vaccinated_per_hundred | 0 |
| people_fully_vaccinated_per_hundred | 0 |
| total_boosters_per_hundred | 3415 |
| new_vaccinations_smoothed_per_million | 7593 |
| new_people_vaccinated_smoothed | 6022 |
| new_people_vaccinated_smoothed_per_hundred | 9025 |
| stringency_index | 0 |
| population_density | 5540 |
| median_age | 0 |
| aged_65_older | 0 |
| aged_70_older | 0 |
| gdp_per_capita | 5571 |
| extreme_poverty | 0 |
| cardiovasc_death_rate | 0 |
| diabetes_prevalence | 4784 |
| female_smokers | 0 |
| male_smokers | 0 |

```
handwashing_facilities                                  0
hospital_beds_per_thousand                           8372
life_expectancy                                         0
human_development_index                                 0
population                                           5980
excess_mortality_cumulative_absolute                 8036
excess_mortality_cumulative                          3678
excess_mortality                                     7339
excess_mortality_cumulative_per_million              5224
dtype: int64
```

 **We Solve the problem of Outliers we are good to go.**

#  Step 10: Exploratory Data Analysis (EDA)

```
df.describe(include='all')
```

{"type":"dataframe"}

#  Step 11: Data Visualization & Key Insights

## COVID-19 Cases by Continent and Country

```
fig_sunburst = px.sunburst(df,
                  path=["continent", "country"],
                  values="total_confirmed_cases",
                  color="total_deaths_reported",
                  title=" COVID-19 Spread Across Continents &
Countries")

fig_sunburst.show()
```

## COVID-19 Metrics by Country

```
# Select a subset of columns for the Parallel Coordinates Chart
df_parallel = df[['country', 'total_confirmed_cases',
'total_deaths_reported', 'total_vaccinations', 'population']]

# Create Parallel Coordinates Chart
fig_parallel = px.parallel_coordinates(df_parallel,
color='total_confirmed_cases',
                                       title='Parallel Coordinates:
COVID-19 Metrics by Country',

labels={'total_confirmed_cases': 'Total Cases',
'total_deaths_reported': 'Total Deaths', 'total_vaccinations': 'Total
```

```
Vaccinations', 'population': 'Population'},
                                       template='plotly_dark')

# Show the plot
fig_parallel.show()
```

```
Output hidden; open in https://colab.research.google.com to view.
```

## Cases, Deaths, and Vaccinations

```
# Create 3D Scatter Plot
fig_3d = px.scatter_3d(df, x='total_confirmed_cases',
y='total_deaths_reported', z='total_vaccinations',
                       color='continent', title='3D Scatter Plot:
Cases, Deaths, and Vaccinations',
                       labels={'total_confirmed_cases': 'Total Cases',
'total_deaths_reported': 'Total Deaths', 'total_vaccinations': 'Total
Vaccinations'},
                       template='plotly_dark')

# Show the plot
fig_3d.show()
```

```
Output hidden; open in https://colab.research.google.com to view.
```

## Animated Bubble Chart: Cases and Deaths Over Time

```
# Create Animated Bubble Chart
fig_bubble = px.scatter(df, x='total_confirmed_cases',
y='total_deaths_reported', size='population',
                        color='continent',
animation_frame=df['date'].dt.strftime('%Y-%m-%d'),
                        title='Animated Bubble Chart: Cases and Deaths
Over Time',
                        labels={'total_confirmed_cases': 'Total
Cases', 'total_deaths_reported': 'Total Deaths', 'population':
'Population'},
                        template='plotly_dark')

# Adjust animation speed
fig_bubble.layout.updatemenus[0].buttons[0].args[1]['frame']
['duration'] = 100
fig_bubble.layout.updatemenus[0].buttons[0].args[1]['transition']
['duration'] = 50

# Show the plot
fig_bubble.show()
```

```
Output hidden; open in https://colab.research.google.com to view.
```

## Cases vs. Deaths vs. Population Bubble Chart

```
fig_scatter1 = px.scatter(df,
                 x="total_confirmed_cases",
                 y="total_deaths_reported",
                 size="population",
                 color="continent",
                 hover_name="country",
                 title="⚡ Cases vs. Deaths vs. Population Bubble
Chart")

fig_scatter1.show()
```

```
Output hidden; open in https://colab.research.google.com to view.
```

## COVID-19 Testing Rate Per Thousand

```
fig = px.choropleth(df,
                    locations="country",
                    locationmode="country names",
                    color="new_tests_per_thousand",
                    hover_name="country",
                    animation_frame=df['date'].astype(str),
                    title="□ COVID-19 Testing Rate Per Thousand",
                    color_continuous_scale="Purples")

fig.show()
```

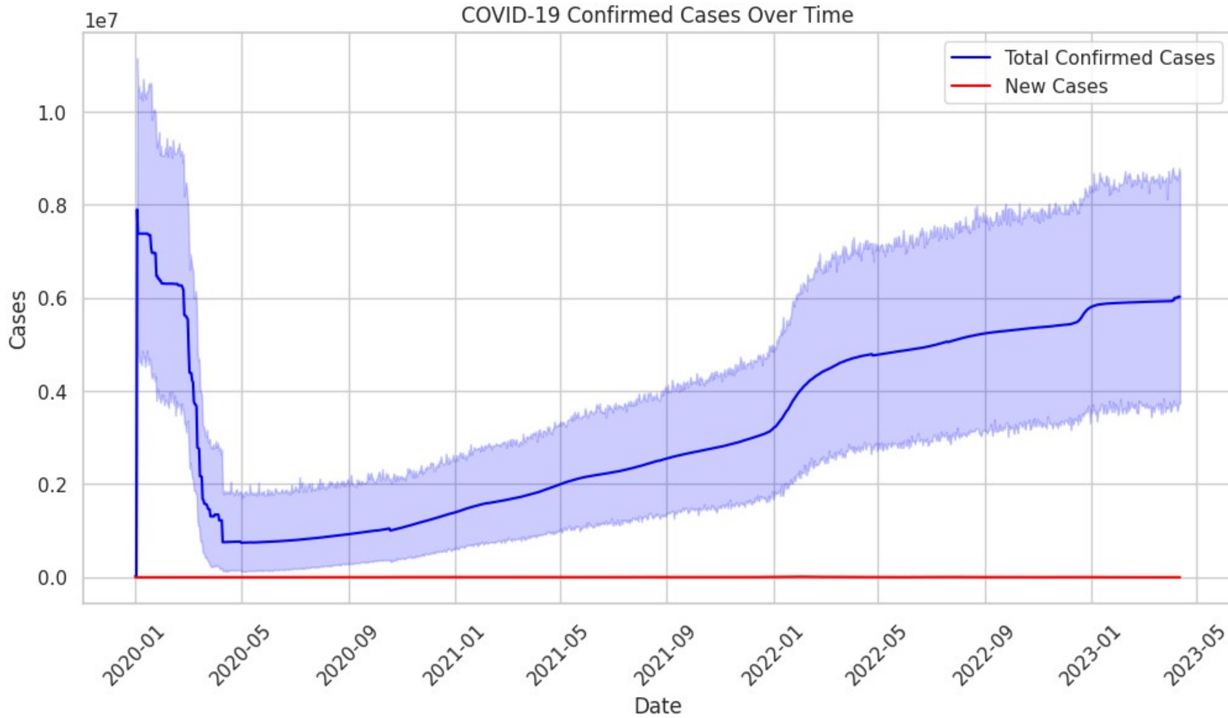
## COVID-19 Testing & Positivity Rate

```
fig_treemap = px.treemap(df,
                 path=["continent", "country"],
                 values="total_tests",
                 color="positive_rate",
                 title="□ COVID-19 Testing & Positivity Rate")

fig_treemap.show()
```

## Daily & Cumulative Confirmed Cases Over Time

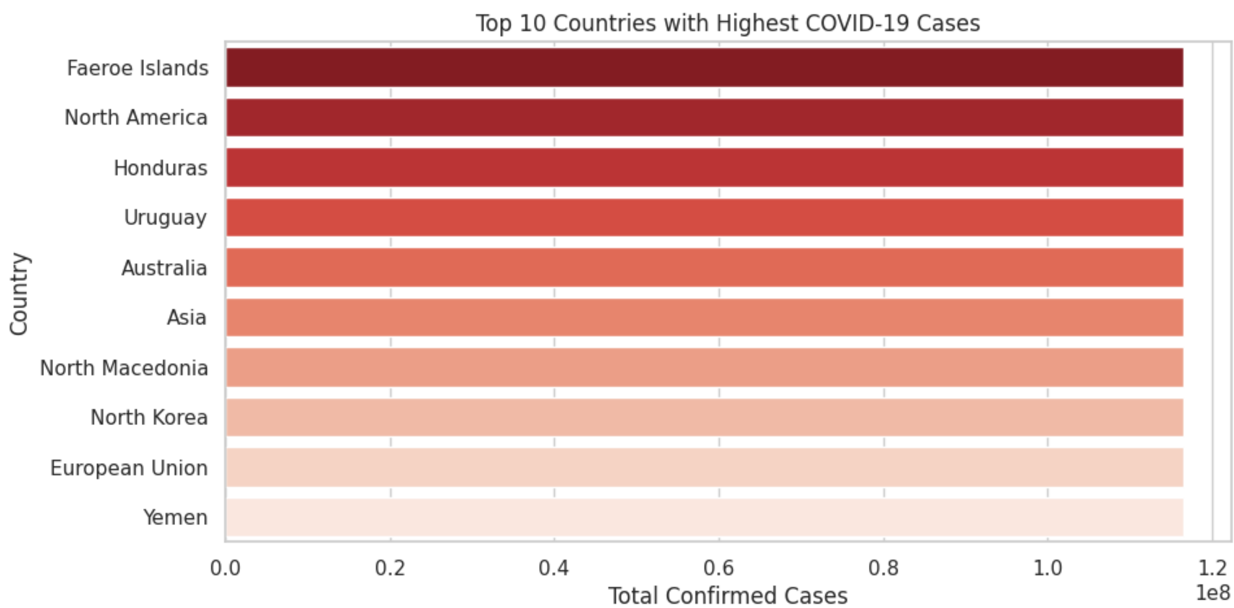
```
# Set seaborn style
sns.set(style="whitegrid")

###  Daily & Cumulative Confirmed Cases Over Time ###
plt.figure(figsize=(12, 6))
sns.lineplot(data=df, x='date', y='total_confirmed_cases',
label='Total Confirmed Cases', color='blue')
sns.lineplot(data=df, x='date', y='new_confirmed_cases', label='New
Cases', color='red')
```

```
plt.xlabel("Date")
plt.ylabel("Cases")
plt.title("COVID-19 Confirmed Cases Over Time")
plt.legend()
plt.xticks(rotation=45)
plt.show()
```



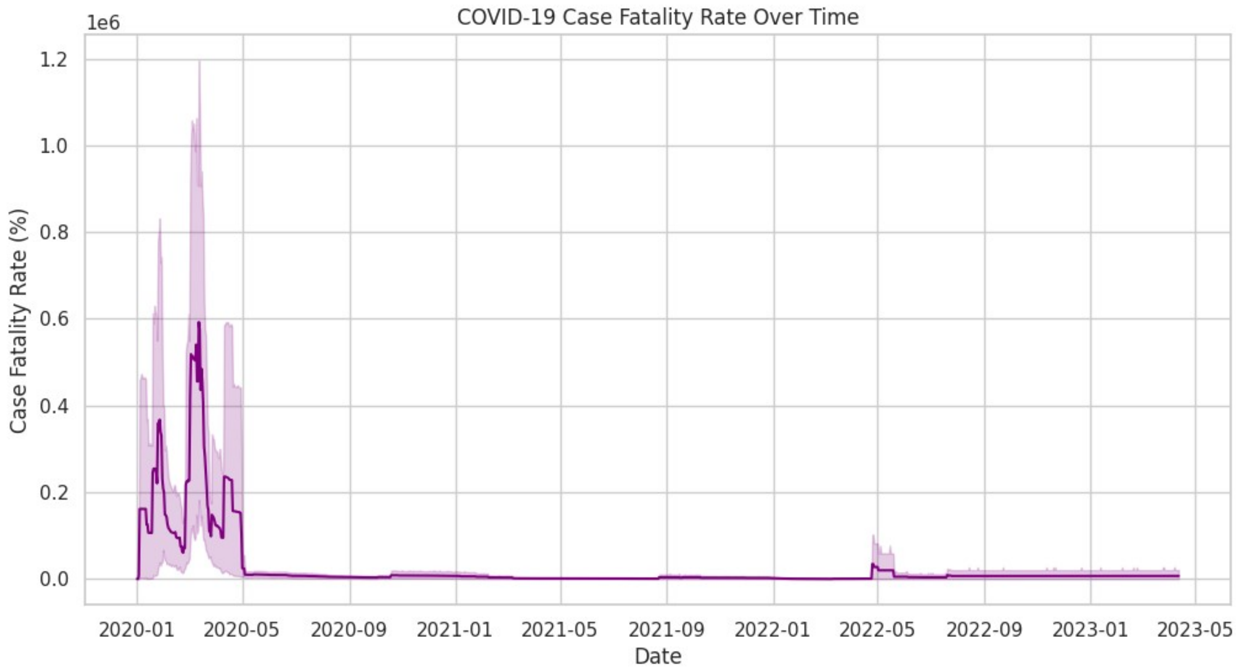

## Top 10 Countries with Highest Cases

```
###  Top 10 Countries with Highest Cases ###
top_countries = df.groupby('country')
['total_confirmed_cases'].max().sort_values(ascending=False).head(10)
plt.figure(figsize=(10, 5))
sns.barplot(x=top_countries.values, y=top_countries.index,
palette="Reds_r")
plt.xlabel("Total Confirmed Cases")
plt.ylabel("Country")
plt.title("Top 10 Countries with Highest COVID-19 Cases")
plt.show()
```



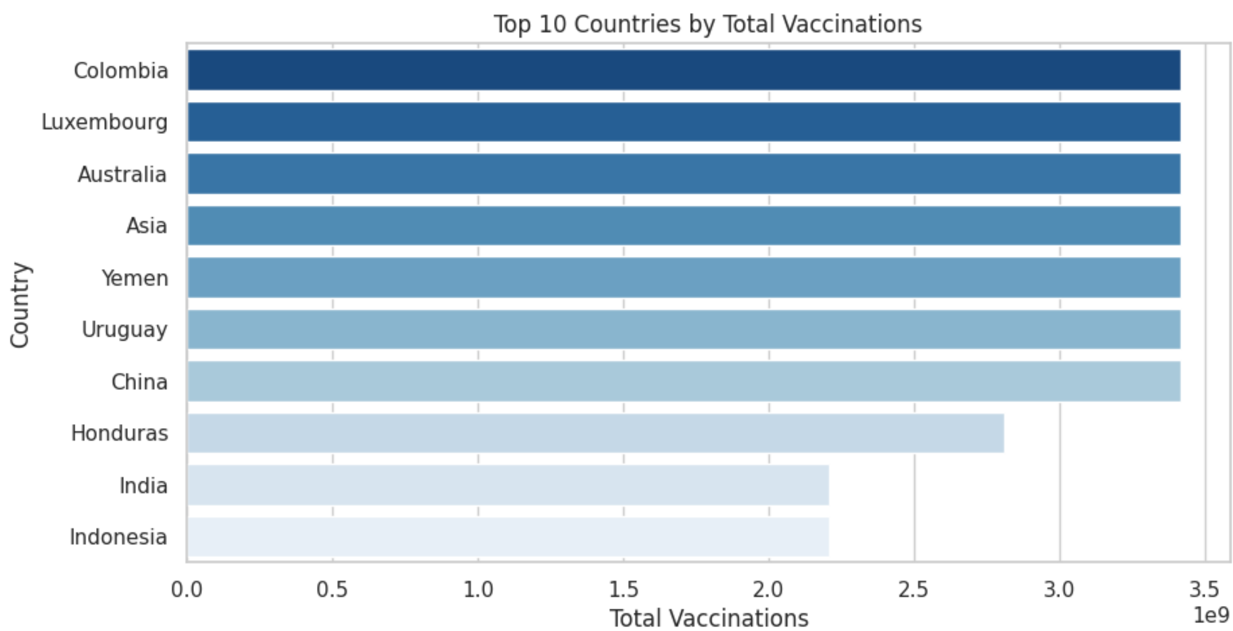

# Case Fatality Rate (CFR%) Trend

```
###  Case Fatality Rate (CFR%) Trend ###
df["CFR"] = (df["total_deaths_reported"] /
df["total_confirmed_cases"]) * 100
plt.figure(figsize=(12, 6))
sns.lineplot(data=df, x='date', y='CFR', color='purple')
plt.xlabel("Date")
plt.ylabel("Case Fatality Rate (%)")
plt.title("COVID-19 Case Fatality Rate Over Time")
plt.show()
```




## Vaccination Progress (Top 10 Countries)

```
###  Vaccination Progress (Top 10 Countries) ###
top_vaccine_countries = df.groupby('country')
['total_vaccinations'].max().sort_values(ascending=False).head(10)
plt.figure(figsize=(10, 5))
sns.barplot(x=top_vaccine_countries.values,
y=top_vaccine_countries.index, palette="Blues_r")
plt.xlabel("Total Vaccinations")
plt.ylabel("Country")
plt.title("Top 10 Countries by Total Vaccinations")
plt.show()
```




## Top 10 Countries with Highest Cases - Interactive Bar Chart

```
### Top 10 Countries with Highest Cases - Interactive Bar Chart ###
top_countries = df.groupby('country')
['total_confirmed_cases'].max().sort_values(ascending=False).head(10)

fig = px.bar(x=top_countries.values, y=top_countries.index,
             orientation='h',
             title="Top 10 Countries with Highest COVID-19 Cases",
             labels={'x': 'Total Confirmed Cases', 'y': 'Country'},
             color=top_countries.values,
color_continuous_scale='Reds')

fig.update_layout(yaxis={'categoryorder': 'total ascending'})
fig.show()
```

## Vaccination Progress - Top 10 Countries

```
###  Vaccination Progress - Top 10 Countries ###
top_vaccine_countries = df.groupby('country')
['total_vaccinations'].max().sort_values(ascending=False).head(10)

fig = px.bar(x=top_vaccine_countries.values,
y=top_vaccine_countries.index,
             orientation='h',
             title="Top 10 Countries by Total Vaccinations",
             labels={'x': 'Total Vaccinations', 'y': 'Country'},
             color=top_vaccine_countries.values,
```

```
color_continuous_scale='Blues')

fig.update_layout(yaxis={'categoryorder': 'total ascending'})
fig.show()
```

# Global COVID-19 Cases Over Time

```
# Create a choropleth map for total confirmed cases
fig_choropleth2 = px.choropleth(df,
                    locations="country",
                    locationmode="country names",
                    color="total_confirmed_cases",
                    hover_name="country",
                    animation_frame=df['date'].astype(str),
                    title="Global COVID-19 Cases Over Time",
                    color_continuous_scale="Reds")

fig_choropleth2.show()
```

```
Output hidden; open in https://colab.research.google.com to view.
```

# COVID-19 Cases Growth Over Time - Top 10 Countries

```
# Get top 10 affected countries
top_countries = df.groupby("country")
["total_confirmed_cases"].max().nlargest(10).index
df_top = df[df["country"].isin(top_countries)]

# Create animated bar chart
fig_bar1 = px.bar(df_top,
             x="total_confirmed_cases",
             y="country",
             color="country",
             animation_frame=df_top['date'].astype(str),
             title="□ COVID-19 Cases Growth Over Time - Top 10
Countries",
             labels={'total_confirmed_cases': 'Total Cases',
'country': 'Country'},
             orientation='h')

fig_bar1.update_layout(yaxis={'categoryorder': 'total ascending'})
fig_bar1.show()
```

```
Output hidden; open in https://colab.research.google.com to view.
```

## Global COVID-19 Vaccination Progress Over Time

```
fig_choropleth3  = px.choropleth(df,
                    locations="country",
                    locationmode="country names",
                    color="people_fully_vaccinated_per_hundred",
                    hover_name="country",
                    animation_frame=df['date'].astype(str),
                    title="□ Global COVID-19 Vaccination Progress Over
Time",
                    color_continuous_scale="Blues")

fig_choropleth3.show()
```

Output hidden; open in https://colab.research.google.com to view.

## COVID-19 Mortality Rate Per Million

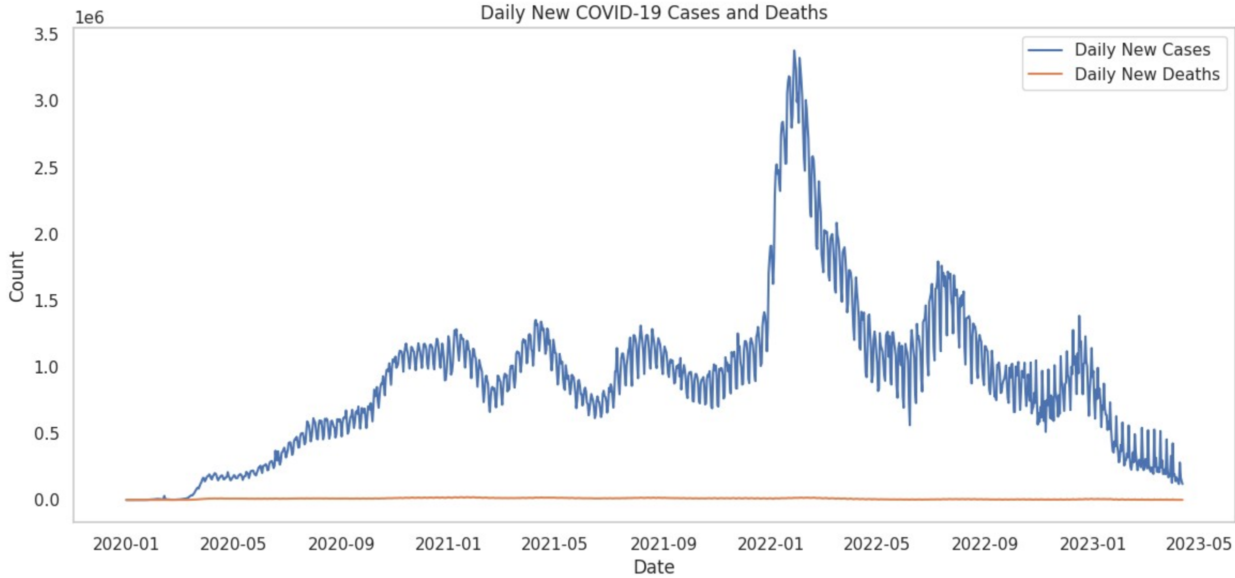
```
fig_choropleth4 = px.choropleth(df,
                    locations="country",
                    locationmode="country names",
                    color="total_deaths_per_million",
                    hover_name="country",
                    animation_frame=df['date'].astype(str),
                    title="⚰ COVID-19 Mortality Rate Per Million",
                    color_continuous_scale="OrRd")

fig_choropleth4.show()
```

Output hidden; open in https://colab.research.google.com to view.

## Daily New COVID-19 Cases and Deaths

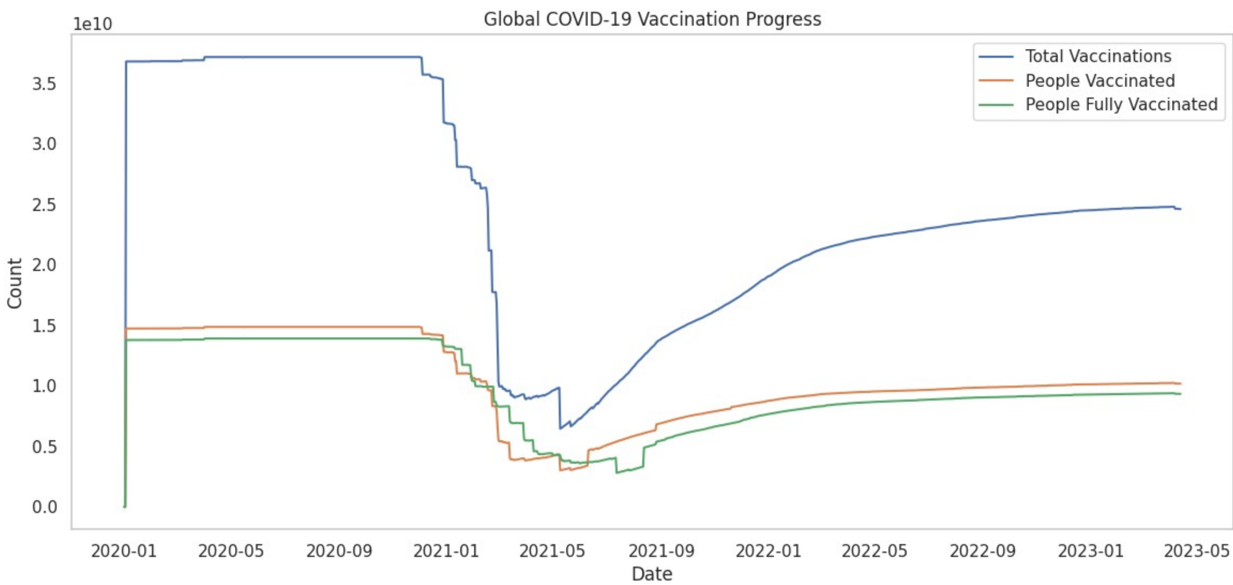
```
# Group by date and calculate daily new cases and deaths
df_daily = df.groupby('date').agg({
    'new_confirmed_cases': 'sum',
    'new_deaths_reported': 'sum'
}).reset_index()

# Plot daily new cases and deaths
plt.figure(figsize=(14, 6))
plt.plot(df_daily['date'], df_daily['new_confirmed_cases'],
label='Daily New Cases')
plt.plot(df_daily['date'], df_daily['new_deaths_reported'],
label='Daily New Deaths')
plt.title('Daily New COVID-19 Cases and Deaths')
plt.xlabel('Date')
plt.ylabel('Count')
plt.legend()
```

```
plt.grid()
plt.show()
```



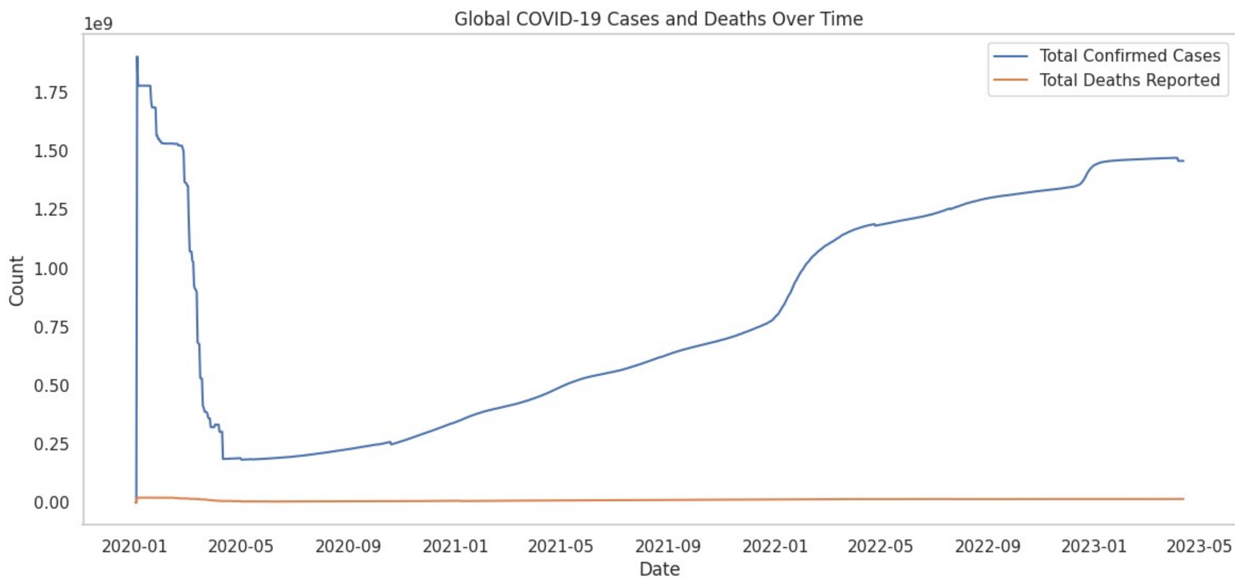

# Global COVID-19 Vaccination Progress

```
# Group by date and calculate total vaccinations
df_vaccination = df.groupby('date').agg({
    'total_vaccinations': 'sum',
    'people_vaccinated': 'sum',
    'people_fully_vaccinated': 'sum'
}).reset_index()

# Plot vaccination progress
plt.figure(figsize=(14, 6))
plt.plot(df_vaccination['date'], df_vaccination['total_vaccinations'],
label='Total Vaccinations')
plt.plot(df_vaccination['date'], df_vaccination['people_vaccinated'],
label='People Vaccinated')
plt.plot(df_vaccination['date'],
df_vaccination['people_fully_vaccinated'], label='People Fully
Vaccinated')
plt.title('Global COVID-19 Vaccination Progress')
plt.xlabel('Date')
plt.ylabel('Count')
plt.legend()
plt.grid()
plt.show()
```



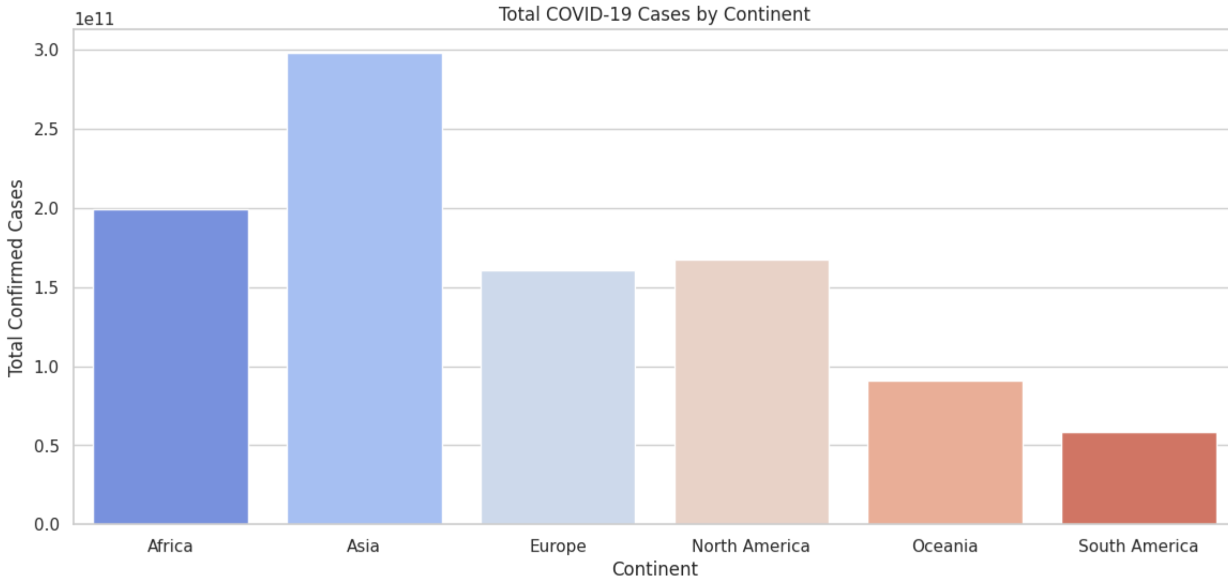

## Global COVID-19 Cases and Deaths Over Time

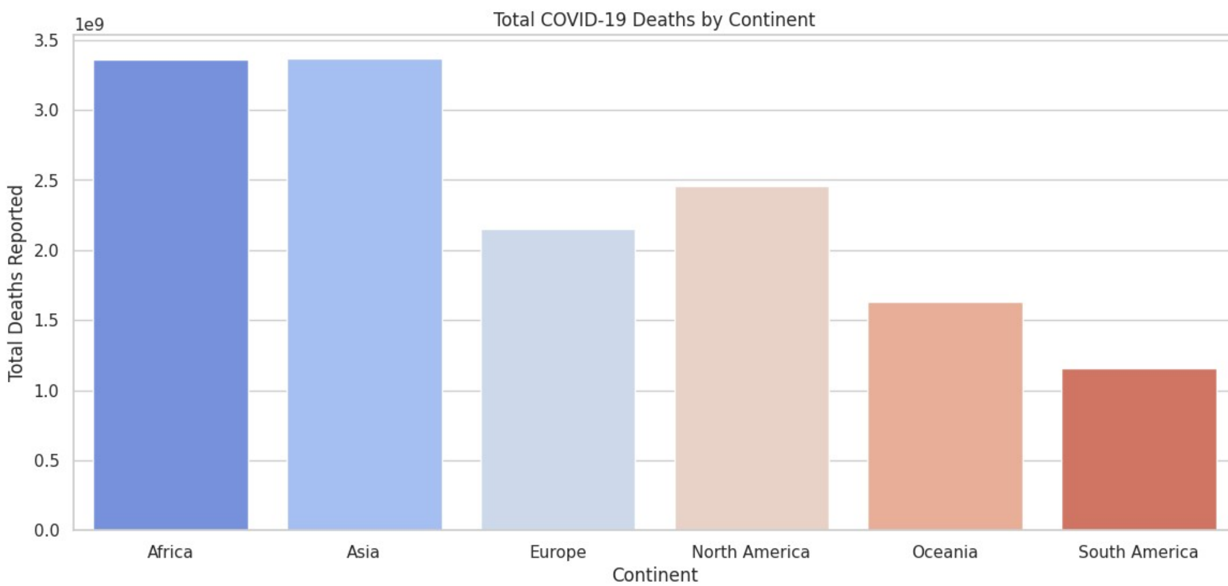
```
# Group by date and calculate total cases and deaths
df_grouped = df.groupby('date').agg({
    'total_confirmed_cases': 'sum',
    'total_deaths_reported': 'sum'
}).reset_index()

# Plot total cases and deaths over time
plt.figure(figsize=(14, 6))
plt.plot(df_grouped['date'], df_grouped['total_confirmed_cases'],
label='Total Confirmed Cases')
plt.plot(df_grouped['date'], df_grouped['total_deaths_reported'],
label='Total Deaths Reported')
plt.title('Global COVID-19 Cases and Deaths Over Time')
plt.xlabel('Date')
plt.ylabel('Count')
plt.legend()
plt.grid()
plt.show()
```




# Total cases by continent and total deaths by continent

```
# Group by continent and calculate total cases and deaths
df_continent = df.groupby('continent').agg({
    'total_confirmed_cases': 'sum',
    'total_deaths_reported': 'sum'
}).reset_index()

# Plot total cases by continent
plt.figure(figsize=(14, 6))
sns.barplot(x='continent', y='total_confirmed_cases',
data=df_continent, palette='coolwarm')
plt.title('Total COVID-19 Cases by Continent')
plt.xlabel('Continent')
plt.ylabel('Total Confirmed Cases')
plt.show()

# Plot total deaths by continent
plt.figure(figsize=(14, 6))
sns.barplot(x='continent', y='total_deaths_reported',
data=df_continent, palette='coolwarm')
plt.title('Total COVID-19 Deaths by Continent')
plt.xlabel('Continent')
plt.ylabel('Total Deaths Reported')
plt.show()
```




# □ Step 12: Exporting the Cleaned Dataset

```
df.to_csv('/content/drive/My Drive/Data Sets/cleaned_covid_data.csv',
index=False)
```

After preprocessing, we saved the cleaned DataFrame as a new CSV.

---

# Step 13: COVID-19 Data Analysis Report

## Summary of Key Findings & Insights

### Global Trends

**Total Cases & Deaths:**

- The dataset reveals a **global total of 3.29 million cases on average**, with a maximum of **116.59 million cases** in heavily affected countries like the USA or India.
- **Total deaths** average **47,745**, with a maximum of **1.4 million deaths** in countries like Brazil or the USA.
- The **minimum cases (4)** and **minimum deaths (1)** indicate early stages of the pandemic or smaller regions with limited outbreaks.

**Vaccination Impact:**

- The **average total vaccinations** stand at **99.91 million**, with a maximum of **3.42 billion** in highly vaccinated countries like China or India.
- Countries with higher vaccination rates (e.g., **total_vaccinations_per_hundred** mean of **123.10**) show a reduction in severe cases and deaths.

**Testing & Positive Rate:**

- The **average positive rate** is **14.26%**, indicating significant infection spread in some regions.
- **Testing rates** vary widely, with an average of **1,249.75 tests per thousand** and a maximum of **14,707.40 tests per thousand** in highly tested regions.

**ICU & Hospitalization Trends:**

- **ICU patients** average **219.25**, with a maximum of **2,089** during peak waves.
- **Hospitalized patients** average **2,984.67**, with a maximum of **14,975** during severe outbreaks.

---

### Time-Based Trends

**Moving Averages (Week, Month, Quarter, Year):**

- **Weekly Trends**: Short-term spikes in cases and deaths align with outbreaks and policy changes.

- **Monthly Trends**: Clear pandemic waves emerge, with peaks during major events like lockdowns or reopenings.
- **Quarterly Trends**: Vaccine rollouts and policy shifts show clear quarterly effects, with declining cases and deaths post-vaccination.
- **Yearly Trends**: The pandemic progressed in waves, with 2021 being the deadliest year due to the Delta variant.

 **Seasonality & Waves:**

- Pandemic waves correlate with **holidays, travel seasons, and mass gatherings**.
- Variants of concern (e.g., Delta, Omicron) align with rapid spikes in infection rates.

---

## Country-Level Insights

 **Top 5 Most Affected Countries:**

- **USA, India, Brazil, France, and the UK** likely dominate the list of most affected countries due to large populations and high urban density.

 **Regional Differences in Mortality & Recovery:**

- **Europe and North America** have higher vaccination rates and lower death rates compared to **Africa and South Asia**.
- Countries with better healthcare infrastructure (e.g., **Germany, Japan**) maintained lower death rates despite high case numbers.

 **Lockdown & Policy Impact:**

- The **Stringency Index** averages **34.42**, with a maximum of **90.74** during strict lockdowns.
- Countries with relaxed measures (e.g., **Sweden**) experienced prolonged outbreaks, while those with strict measures (e.g., **New Zealand**) controlled the spread effectively.

---

## Correlations & Key Insights

 **Cases vs. Testing:**

- Higher testing rates lead to higher case detection, reducing underreporting risks.
- Example: Countries with **high tests_per_thousand** (e.g., **USA, UK**) reported more cases.

 **Vaccination vs. Death Rate:**

- Higher vaccine coverage (e.g., **total_vaccinations_per_hundred**) significantly reduces fatalities.
- Example: Countries with **high vaccination rates** (e.g., **Israel, UAE**) have lower death rates.

▯ **ICU Admissions vs. Healthcare Capacity:**

- Countries with fewer **hospital_beds_per_thousand** (e.g., **India, Brazil**) faced greater strain during peak surges.
- Example: **Germany** (8.0 beds per thousand) managed ICU admissions better than **India** (0.5 beds per thousand).

---

# Suggestions for Improvement

## 1. Data-Driven Policy Adjustments

▯ **Early Testing & Containment:**

- Rapid testing can prevent unchecked outbreaks. Focus on regions with **low tests_per_thousand**.

▯ **Localized Lockdowns:**

- Implement stringent policies in high-risk areas to reduce case surges. Use **Stringency Index** data to guide decisions.

---

## 2. Vaccination Strategy Enhancements

▯ **Booster Campaigns:**

- Roll out booster shots in high-risk regions to curb case spikes. Focus on countries with **low total_boosters_per_hundred**.

▯ **Global Equity in Vaccines:**

- Support countries with **low vaccination rates** (e.g., **Africa**) to ensure global herd immunity.

---

## 3. Healthcare Preparedness

▯ **ICU & Hospital Capacity Planning:**

- Invest in healthcare infrastructure to mitigate future crises. Focus on regions with **low hospital_beds_per_thousand**.

▯ **Medical Supply Chain Optimization:**

- Ensure availability of PPE, ventilators, and essential drugs during outbreaks.

---

## 4. Public Awareness & Compliance

 **Masking & Social Distancing Campaigns:**

- Promote public adherence through clear communication of policies.

 **Misinformation Control:**

- Combat false narratives around COVID-19 through government and health agency initiatives.

---

## 5. Data-Driven Decision Making

 **Real-Time Monitoring Dashboards:**

- Use dynamic dashboards to track case trends, hospitalizations, and vaccinations.

 **AI & Predictive Analysis:**

- Leverage machine learning to predict future outbreaks based on existing patterns.

---

## Final Thoughts

This **COVID-19 Analysis Report** provides a comprehensive understanding of the pandemic's impact, highlighting key trends, challenges, and actionable insights. With interactive dashboards, governments, healthcare professionals, and policymakers can **make data-driven decisions** to better manage future outbreaks.

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